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# Capturing complex hand movements and object interactions using machine learning-powered stretchable smart textile gloves

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Accurate real-time tracking of dexterous hand movements has numerous applications in human–computer interaction, the metaverse, robotics and tele-health. Capturing realistic hand movements is challenging because of the large number of articulations and degrees of freedom. Here we report accurate and dynamic tracking of articulated hand and finger movements using stretchable, washable smart gloves with embedded helical sensor yarns and inertial measurement units. The sensor yarns have a high dynamic range, responding to strains as low as 0.005% and as high as 155%, and show stability during extensive use and washing cycles. We use multi-stage machine learning to report average joint-angle estimation root mean square errors of 1.21° and 1.45° for intra- and inter-participant cross-validation, respectively, matching the accuracy of costly motion-capture cameras without occlusion or field-of-view limitations. We report a data augmentation technique that enhances robustness to noise and variations of sensors. We demonstrate accurate tracking of dexterous hand movements during object interactions, opening new avenues of applications, including accurate typing on a mock paper keyboard, recognition of complex dynamic and static gestures adapted from American Sign Language, and object identification.

Real-time tracking of hand movements[1–10] has notable applications in human–computer interaction, electronic gaming, the metaverse and augmented reality[4,11,12], rehabilitation[4,13–16], sports training, robotics, and tele-surgical applications[4,11]. Fuelled by recent advances in machine learning (ML) and flexible electronics[1–5,12,17], significant progress has been reported for tracking or gesture recognition using both computer vision (CV)[4,18–28] and wearable technologies[2–5,12–14,17,29–38]. Fixed and costly motion-capture camera systems with markers are often used as the gold

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**Table 1 | Comparison of overall performance parameters of the smart textile glove and ML system presented in this work and other related published works**

| | This work | Wen et al.[12] | Luo et al.[1] | Moin et al.[2] | Zhou et al.[3] | Hughes et al.[8] | Glauser et al.[5] |
|---|---|---|---|---|---|---|---|
| Form | Smart textile glove | Smart textile glove | Smart textile apparel | Flexible PCB armband | Smart textile glove | Smart textile gloves | Smart latex glove |
| Sensors | 25 HYSs+2× 9-axis IMUs | 15 triboelectric sensors | 32×32 yarn-based piezoresistive pressure sensors | 64 surface electromyography sensors | 5 yarn-based strain sensors | 120 resistive knit sensors+6 fluidic pressure sensors | 44 capacitive silicon sensor array |
| Output | Fingers+wrist-joint angles | | Tactile feedback | 21 gestures | 11 gestures | Finger-joint angles | Finger-joint angles |
| | Keyboard typing | | | | | Heart rate | |
| | 3D drawing | 50 dynamic gestures | | | | 10 handwritten letters | |
| | 50 dynamic gestures | +20 sentences | | | | Stiffness | |
| | 48 static gestures | | | | | Temperature | |
| | 34 objects | | | | | 30 objects | |
| | Force | | | | | Force | |
| Robustness | ✓ | ✗ | ✓ | ✗ | ✗ | ✗ | ✗ |
| Washability | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |
| Wireless | ✓ | ✗ | ✗ | ✓ | ✓ | ✗ | ✗ |
| Sampling rate (Hz) | 20 | 600 | 14 | 1,000 | 500 | 16 | 600 |
| ML response time (ms) | 5 | 83 | 100 | 250 | 1,000 | Not mentioned | 125 |
| Offline personalized training time | 1 min | Not mentioned | Not mentioned | Not mentioned | 2 min | MLP: ~40 min | 2 h |
| No GPU needed to calibrate | ✓ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ |
| Finger-joint-angle estimation error | Intra: 1.24° | ✗ | ✗ | ✗ | ✗ | MLP: 6.40° | Intra: 5.8° (2 h), 6.5° (20 min) |
| | Inter: 1.45° | | | | | LSTM: 4.78° | Inter: 6.2° (2 h), 7.6° (1 min) |
| Static gesture recognition accuracy | Intra: 97.81% | Not mentioned | ✗ | Intra: 92.87% | Threefold cross-validation | Not mentioned | Not mentioned |
| | Inter: 94.60% | | | Inter: 84.53% | 98.63% | | |
| Dynamic gesture recognition accuracy | Intra: 97.31% | 50 words: 91.3% | ✗ | ✗ | ✗ | Not mentioned | Not mentioned |
| | Inter: 94.05% | 20 sentences: 95.0% | | | | | |
| Typing accuracy | Intra: 97.8% | Not mentioned | ✗ | ✗ | ✗ | ✗ | ✗ |
| Object-detection accuracy | Intra: 95.02% | ✗ | ✗ | ✗ | ✗ | MLP: 99.69% | ✗ |
| | Inter: 90.20% | | | | | LSTM: 98.18% | |
| Wrist-angle estimation error | Intra: 2.09° | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |
| | Inter: 2.45° | | | | | | |

Note that different advanced wearable systems reported in literature function based on different physiological information (for example, electromyography and tactile), sensing modalities and form factors and can be used in variety of applications, which may not be fair to compare based on only performance parameters. MLP, multi-layer perceptron; GPU, graphics processing unit.

standard for detailed and articulated hand and finger tracking. CV-based solutions using one or more cameras either attached to the user's headset[6,15,27] or placed in a specific location[15,28] are demonstrated as lower-cost consumer solutions. Both motion-capture and CV-based technologies are spatially limited to the field of view of cameras and face major challenges due to occlusion by objects, hands or other body parts, poor lighting, and background noise[4,11,15]. Current wearable technologies are primarily used for gesture recognition in the form of gloves[1,3,5,12,39], wrist bands[25] or arm sleeves[2,17]. Researchers have reported integration of different sensors, including surface electromyography electrodes[2,17], pressure sensors[3,5–10,40], inertial measurement units (IMUs)[4], magnetic sensors[4] and optical sensors[28,41]. However, most of these wearable devices are used to detect only specific gestures with limited accuracy and have not addressed challenges related to reliability, accuracy and washability of the device[1,4]. Current solutions have strict requirements for placement of sensors directly on a user's hand and do not address variations in electrical and mechanical properties of the sensors and fit to the users[1,2,4,17]. These factors, in addition to lack of washing or sanitization methods, lead to limited practical usability and accuracy of these solutions[1,4,42].

In this Article, we report a dynamic and accurate tracking of hand movements, articulated for all finger and wrist joints, using stretchable, wireless and washable smart textile gloves embedded with stretchable helical sensor yarns (HSYs), IMUs and stretchy interconnects. Using our multi-stage ML algorithms, we report an average joint-angle estimation root mean square error (r.m.s.e.) of 1.21° and 1.45° for intra- and inter-participant cross-validation, respectively, going well beyond the accuracy of published wearable devices and CV systems[1–5]. Our results indicate that the proposed smart gloves and ML algorithms provide a tool for learning dexterous hand and finger movements that goes beyond conventional gesture recognition and compete in accuracy with costly motion-capture systems without limitations of field of view, long set-up times (for example, camera calibration time, marker set-up time), and sensitivity to occlusion and image noise that are highly prevalent in hand-tracking applications. Indebted to the high dynamic range and reliability of the HSYs in response to stretches and pressures at the tip of the fingers as well as our ML algorithms, we demonstrate reliable tracking of complex hand and finger movements during interaction with objects, which is not practical in camera systems due to occlusion by objects and fingers. We also demonstrate a data augmentation technique that increases the robustness of our results by two times in the presence of sizable variations in the performance of the sensors and their fit to the participant. On the basis of these results, we demonstrate complex potential applications such as dynamic tracking of hand and finger movements, accurate (97.80%) typing on a mock paper keyboard, highly accurate dynamic gesture recognition (94.05% and 97.31% inter- and intra-participant cross-validation accuracy, respectively, for 50 gestures), static gesture recognition (94.60% and 97.81% inter- and intra-participant cross-validation accuracy, respectively, for 48 gestures), and object recognition from grasp patterns (90.20% and 95.02% inter- and intra-participant cross-validation accuracy, respectively, for 34 objects). Table 1 summarizes the overall performance parameters of our system compared with other published works. We believe that our stretchable smart textile glove and ML algorithms can unlock new avenues in human–computer interaction, movement and therapy assessment in remote health, and applications in animation and the metaverse for learning dexterous human-hand functions and interactions with objects and surfaces.

## Smart textile gloves

Figure 1a shows the main functionality of the smart textile glove, including joint-angle estimation and detecting the grasp pressure in interaction with objects. Figure 1b shows a photograph of our glove with a schematic overlay showing the embedded HSYs, specifically located at each finger joint, the tip of each finger, the palm and the thumb joints. Wavy three-dimensional (3D) stretchable interconnects connecting all sensors to a wireless processing board with a rechargeable battery as well as two nine-axis degrees of freedom IMUs are integrated on the dorsal side of the hand and within the textile on top of the forearm. As discussed in the next section, the HSYs are integrated on the surface of a stretchable internal glove fabric and connected using wavy insulated stretchable interconnects. The gloves are then covered with another layer of stretchable fabric and sewn together to provide reliable, durable and accurate operation. The HSYs can detect local deformations including stretches and pressures in the fabric caused by the movements of joints and fingers or forces when interacting with objects or when pressed against one's own hand tissue (Fig. 1c). The two nine degrees of freedom IMUs enable highly accurate tracking of wrist-joint quaternion angles. Figure 1d shows a schematic diagram of the system model. We used a custom-made wireless processing board (Texavie) that consists of amplifiers, multiplexers, analogue-to-digital converters (ADCs), microprocessors and Bluetooth low-energy (BLE) wireless transmission modules that interact with all the sensors and IMUs and transmit data to an external receiver. We use an iOS mobile app (Texavie) on iPhone or iPad (Apple), or a PC as a data gate and use an ML data pipeline that performs all data processing and ML models, including GlovePose model, 3D visualization (Fig. 1e), contact detection, and gesture- and object-detection algorithms. An ML-based hand-joint-angle estimation algorithm is developed as GlovePoseML model, which was validated against a motion-capture system, as explained in the next sections.

## Helical sensor yarns

Figure 2a illustrates the structure of the HSY, which is critical to accurate performance of the smart glove for detection of finger and hand movements used in the ML models. The stretchable HSYs consist of an elastic core yarn wrapped with metal-coated nanofibres (NFs) in helical form as shown in the figure. The HSY structure is completed with a final protective coating of polydimethylsiloxane (PDMS) and silicone as the outer shell. PDMS can penetrate deeply into the NF mesh and bond the NFs together during bending, pressing and stretching, providing high durability and dynamic range for the piezoresistive HSYs. Silicone elastomer is then applied over the yarn to attach the sensors to the fabric substrate and to protect the PDMS layer as well as providing stable electric insulation during washing. All the processing steps are scalable to roll-to-roll manufacturing including sputtering for economically viable fabrication. Figure 2b,c shows the structure of a typical HSY before and after PDMS and elastomer coatings. The NFs form a helical porous layer around the elastic core yarn. With the PDMS matrix binding the fibres together, the contact area of the metalized NFs changes during the external stretching/pressing–releasing cycles, resulting in a change of resistance in response to strain[43,44]. The helical structure of the NFs ensures that the HSYs maintain linear responses to a wide range of strains and stresses. Low-temperature curing of the PDMS layer (~120 μm) was employed for higher tensile strength and stretchability[45], while the elastomer shell further improves the tensile strength of the yarns. We conducted multiple experiments on different parameters of PDMS coating, such as thickness, curing time and temperature, and different mould shapes and materials to achieve the optimum HSY dynamic range, sensitivity and durability. We found that an optimized thickness of 120–200 μm PDMS thickness and a 40–60 μm core HSY provides optimum performance. The HSYs can reach up to 155% stretchability and ultra-flexibility as shown in Fig. 2c.

Figure 2d depicts the sensor response, defined as change of the yarn resistance to its original value, for the HSYs to a uniaxial strain ranging from 0.005% to 155% at a frequency of 1 Hz, highlighting the exceptional dynamic range and uniformity of the HSYs. As the inset shows, the HSYs can accurately respond to external strains as low as 0.005%, which outperforms the performance of other published wearable sensors to the best of our knowledge, while still maintaining the response at 155% strain[46]. In addition, the HSYs respond to strains along both longitudinal and vertical directions. Compressive pressure tests on the sensors are illustrated in Extended Data Fig. 1c and demonstrate that the sensors can respond to pressure levels as low as 1.7 kPa and up to 1.2 MPa. The exceptional dynamic range, sensitivity, low hysteresis, high linearity and reliability of the HSYs are key for achieving accurate response in the smart glove and tracking of small joint movements, where the stretch in glove fabric is expected to be less than 120%. Figure 2e shows the time-dependent response of the HSYs to uniaxial straining cycles with different maximum magnitude of 1%, 2%, 5%, 10%, 20%, 40%, 80% and 140%. The dependence of the output signals on the frequency of the strain cycle is shown in Fig. 2f. The HSY

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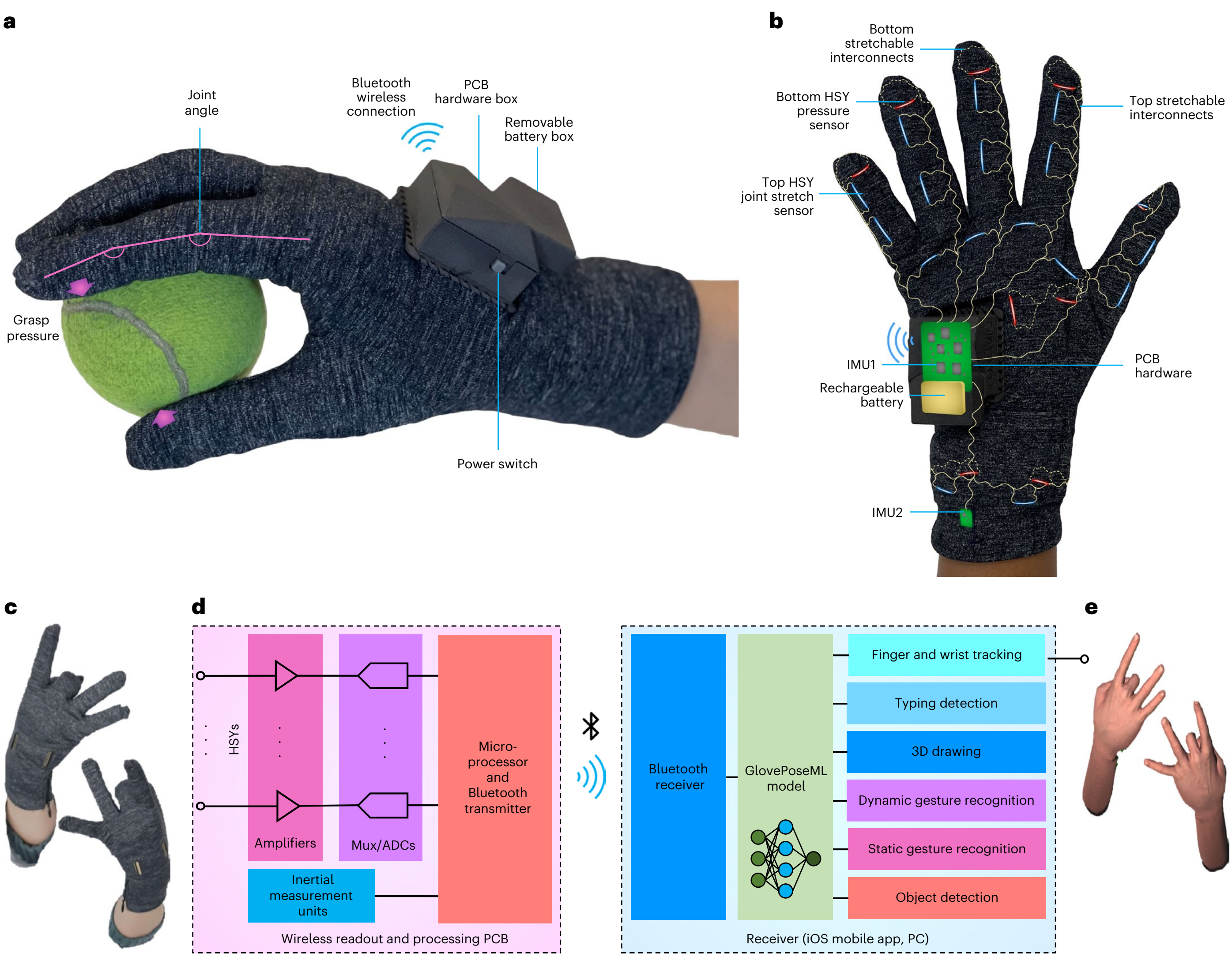


**Fig. 1 | Smart textile glove. a**, Photograph of the smart textile glove demonstrating its functionality in capturing joint angles and grasp pressure in interaction with objects. **b**, Photograph of the glove with X-ray schematic showing embedded HSYs (blue, top; red, bottom), 3D stretchy interconnects (gold lines; solid, top; dashed, bottom), the PCB including the first IMU1, and other readout and Bluetooth hardware, a battery box, and the second embedded IMU2, located just above the wrist. **c**, User wearing a glove pair showing a complex gesture. **d**, Schematic block diagram of how multiple HSYs and IMUs are connected to PCB hardware, including amplifiers, ADCs, multiplexers (Mux), a microprocessor and BLE transmitter, and subsequently to an iOS mobile app or a PC that receives the data and passes it to the GlovePoseML model and demonstration apps for different applications. **e**, Visualization of a user's complex hand gesture estimated by an ML algorithm that dynamically follows the movements.

sensors demonstrate accurate sensing performance under bending radius of down to 2.5 mm and repeated pressure from sharp objects such as a 0.5 mm stainless steel plate and a 0.6-mm-diameter plastic pipette tip up to about 2.1 MPa.

The response time of the HSYs to external strain stimuli is also characterized to evaluate the response time of the system. Figure 2g shows exceptional synchronization of the stimuli and output signals, illustrating the possibility for real-time dynamic motion tracking. The durability of the sensors was studied by conducting a 14-h-long overnight test at the strain level of 10% and 0.7 Hz frequency (Fig. 2h). The sensitivity of the response to water is also shown in Fig. 2i, comparing the response in air and when fully immersed in water. The sensors show less than 10% degradation in sensitivity over the entire range of strains. The sensors are integrated on the fabric and stability and washability testing was undertaken over various day-by-day laundry tests, as shown in Fig. 2j. The sensors were first immersed in water, at various temperatures for different times ranging from 5 min to 50 min with magnetic stirring at different speeds (80 rpm, 160 rpm or 400 rpm). The fabric and sensors were dried in air completely before the measurements of sensitivity. Similarly, detergent and softener were added to the water, and the response of the sensors was recorded after they were dried. The sensors show excellent stability after the simulated drying cycles. In the end, we conducted real washing and drying tests by putting the sensors into the laundry machines together with a full load of other clothes, to study the performance of the sensors in actual daily usage scenarios, demonstrating stability during laundry cycles. HSYs demonstrate superior specifications in comparison to the state-of-the art strain sensors, including broad dynamic range[40,47,48], fast response time[49–51], durability in extensive cycling[46] and excellent washability[40,52]. In addition to washability for HSY embedded fabric, the enclosure box used for the printed circuit board (PCB) hardware is designed with embedded custom-made waterproofing rubber flanges that block water leakage in normal washing conditions, and the PCB is coated with a protective coating and placed in a plastic pouch. This makes the entire glove system washable after removal of the rechargeable battery, tested for over tens of cycles of washing and drying.

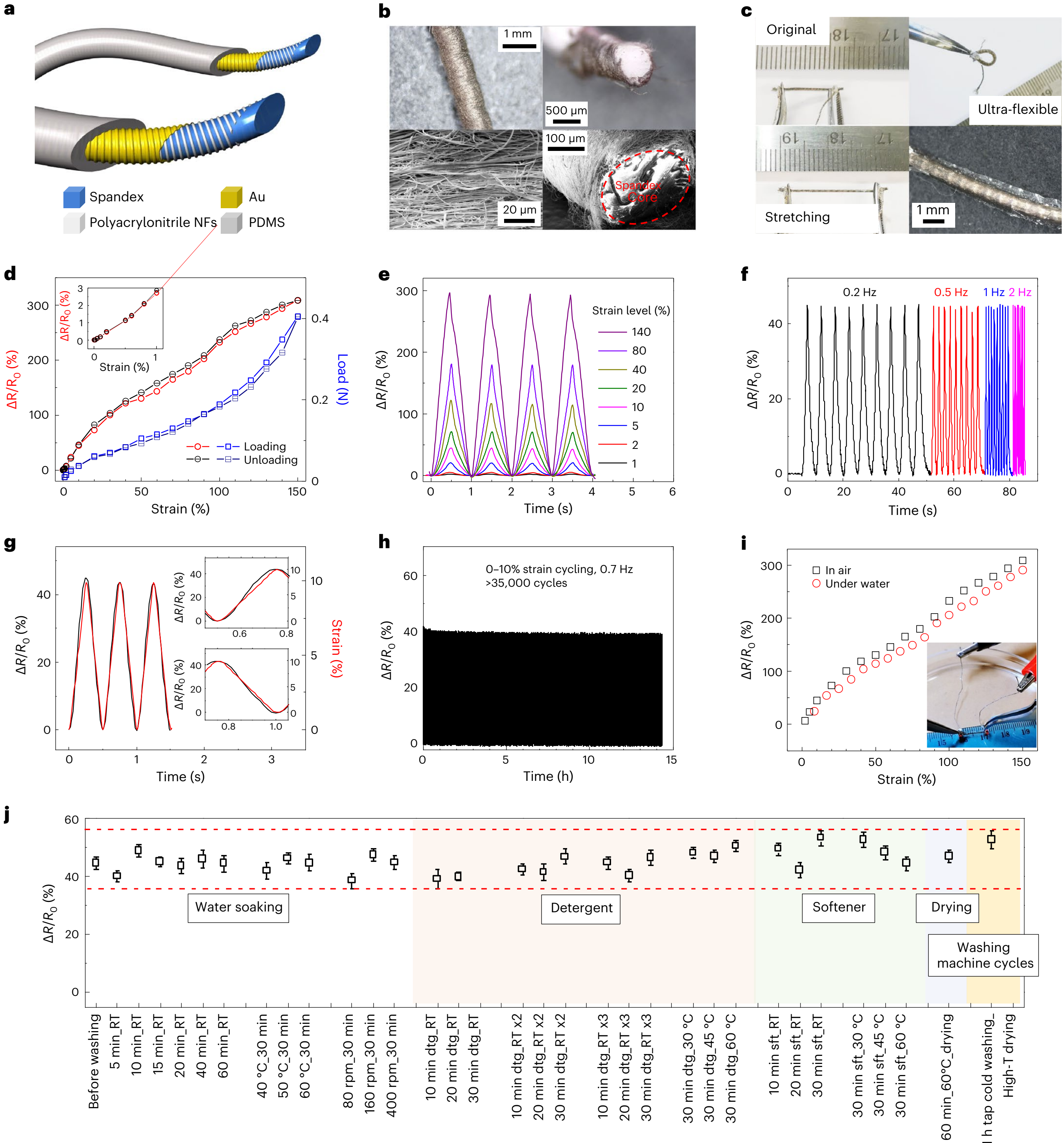


**Fig. 2 | Helical sensor yarns. a**, Schematic of HSYs showing coaxial structure with elastic spandex core, wrapped with helical metal-coated NFs, and encapsulation elastomer shell. **b**, Microscope and scanning electron microscopy photomicrographs of HSYs before shell coating. **c**, Photographs of HSYs with shell and contact electrodes. **d**, Measurement of sensitivity $\Delta R/R_0$, where $\Delta R$ is the change in initial resistance $R_0$, in response to different tensile strains and loads, during loading and unloading, showing exceptional sensitivity down to 0.005% strain and minimal hysteresis. Inset: sensitivity for strains <1.0%. The data points present the mean value of 20 samples. The error bars derived from standard deviation are too small to show on the figure. **d,** Time-dependent sensitivity of resistance to cyclic stretching to a maximum value ranging from 1% to 140%. **f**, Changes in resistance response of HSYs to strains up to a maximum of 10% strain with various frequencies 0.2–2 Hz. **g**, Sensor response accurately following changes in strain from 0 to 10%. Insets: zoomed-in views of loading and unloading cycles. **h**, Mechanical durability test for up to 14 h continuous stretch–release cycles. **i**, Comparison of sensor response in air and underwater. Inset: photograph of sensor during underwater tests. The data points present the mean value of 20 samples. The error bars derived from standard deviation are too low to show. **j**, Durability sensor response in the smart textile glove during various laundry, washing and drying cycles. dtg, detergent; sft, softener; RT, room temperature; RT x2, second RT washing cycle; rt x3, third RT washing cycle; T, temperature. Error bars are standard deviation for nine devices undergoing same experiments. Dashed red lines are the observed minimum and maximum limits during these experiments.

## Machine learning model

In this work, we developed an ML model that can dynamically estimate joint angles for all finger joints and the wrist with high accuracy. This ML model (GlovePoseML) is the core of the algorithm, which is supplemented with other neural network-based models in the output to tune the response for specific applications and demonstrations such as keyboard and object detection. To develop and train GlovePoseML model, we collected a large dataset (over 3,000,000 frames with a sampling rate of 20 Hz) from our smart glove with attached markers for the gold-standard motion-capture system from five participants with different hand sizes. The participants perform complex hand transient movements collected for various tasks, including random finger movements, grasping objects and switching between different gestures (for example, making fist or paper). As shown in Fig. 3f, our GlovePoseML model consists of a two-layer stacked recurrent neural network-based bidirectional long short-term memory (Bi-LSTM) followed by two fully connected (FC) layers and two activation layers. We use this regression architecture for estimating hand-joint angles and tactile information from incoming data. Our model uses a 2 s history of normalized sensor and motion-capture data as input for training. For inter-participant cross-validation, we select one user's data as the testing dataset and the others' as the training dataset, repeat this step for all users, and report the average results. For intra-participant cross-validation, we performed tenfold cross-validation over each user's data and report the average value. With the core model trained, test data can be fed to the model and estimated joint angles are sent through a WebSocket for 3D visualization using Unity software. Figure 3a shows a linear plot comparison of angles estimated by our ML model for the test data and the motion-capture gold standard for different finger-joint flexion or abduction (if applicable) and wrist-joint flexion, abduction and supination. Figure 3c,d shows the inter- and intra-participant cross-validation results for $R^2$ and r.m.s.e. of the joint angles, respectively, showing the model's accuracy for each joint. More detailed results for the accuracy values can be found in Supplementary Table 1. In particular, the results demonstrate a high accuracy of 1.45° (minimum 0.69°, maximum 2.71°) and 1.21° (minimum 0.36°, maximum 2.26°) r.m.s.e. for inter- and intra-participant cross-validation results, respectively. We have demonstrated that our system operates with accuracy in scenarios where current CV methods and wearable devices provide limited or low-accuracy results, such as random finger movement in complex gestures (Fig. 3b, top), visual occlusion by objects (Fig. 3b, middle) and dark conditions (Fig. 3b, bottom), shown in detail in our demonstration video. Such high accuracy values arise from several factors, including the stability and dynamic range of our HSYs and reliable positioning of the HSYs on the joints in stretchable smart gloves, leading to similar responses and trends among different participants and different sessions and accurate performance of our ML algorithm. Our results indicate that our smart glove and ML model can potentially replace high-end, bulky and expensive motion-capture systems and is not prone to common issues associated with these systems, including losing track of markers due to occlusion, light noise or marker swap.

## Data augmentation technique

In practical applications, the smart gloves and the embedded HSYs have small but unavoidable variations in the values of resistances and stretch sensitivities and how they fit to different participants or the same participant in different sessions. By drawing inspiration from nature that adapts to changes in the external environment, we developed a data augmentation technique to enhance robustness to the small variations of the HSYs, their placement and fit to the participants. As shown in Fig. 3e, the pre-training algorithm uses data augmentation to duplicate (1) the original training data and apply some expected variations to the original data, including (2) random sensor amplitude scaling or d.c. shift, (3) random sensor masking or (4) additive noise. Then we use the augmented dataset to train GlovePoseML model to enhance robustness. Figure 3g,h shows the average $R^2$ and r.m.s.e., respectively, for the original model as well as that incorporating the data augmentation technique. These results highlight that the proposed pre-training is highly effective in improving the accuracy (~2× reduction in average r.m.s.e.) of the system in the presence of unavoidable variations. This step enables our ML model to capture invariant representations of the movement accurately and also reduces the necessity for excessive calibration and retraining in the case of variations in the performance or fit. By developing this pre-training method (Supplementary Algorithm 1), we significantly improve the robustness of our smart glove compared with published work that is typically made in the lab and mostly tested by attaching sensors to joints of participants in a highly controlled and unrepeatable fashion[3,4].

## Application examples

We rely on our dexterous hand movements to perform everyday tasks, from manipulating objects, using a computer or communicating with each other[4]. To demonstrate our work's potential applications, we demonstrate six examples of our smart gloves and ML system for capturing complex real-time hand and finger poses and movements, typing on a mock keyboard, tracking movements in air, interactions with and grasping of objects, and dynamic and static gesture recognition. These applications demonstrate how achieving dynamic accuracy can open up applications for our smart gloves and ML system.

For the first example, we report dynamic (ML response time of ~5 ms in processing of data on a regular PC with Intel Core i9-9900K CPU @ 3.5 GHz, and 32 GB RAM DDR4) articulated tracking of finger movements in complex dynamic gestures as shown in the demonstration video. As seen, the system can follow finger movements and all finger and wrist joints with a high level of accuracy. When the model is trained, the accuracy is not affected in dark conditions or when the hand is grasping a ball, which creates visual occlusion for the motion-capture cameras making them practically ineffective. This impressive dynamic performance and accuracy is a substantial improvement compared with previously reported works in the literature and can find applications in animation, the metaverse and tele-operation (a summary of the performance of our glove system versus other similar systems can be found in Table 1).

For the second example, we demonstrate typing on a mock keyboard printed on a piece of paper laid on a table, which serves as a guide for the user during typing (Fig. 4a). We employed the GlovePoseML model trained for both hands and then added two FC layers (Fig. 4c) along with a click-detection algorithm running on the response of HSYs at the tip of finger as shown in Fig. 4d. Inter-session cross-validation of the system's accuracy is shown by different colours in Fig. 4b and shows an average 97.80% accuracy for the prediction of the typed letters. As shown in the demonstration video, the accuracy can support complex keyboard functions such as holding shift for capital letters not demonstrated in previous works. Detailed accuracy results can be found in Extended Data Fig. 3.

For the third example, we considered 3D drawing in the air using the glove as shown in Fig. 4e. In this scenario, users can manoeuvre over an iOS mobile interface application, as shown in the demonstration video, and use their wrist movements to draw with different colours selected based on which finger pinches the thumb. We reported 2.48°, 2.34° and 2.54° errors for estimating the angle of wrist flexion or extension (Fig. 4f), abduction or adduction (Fig. 4g), and supination and pronation (Fig. 4h), respectively, when compared with the motion-capture system.

For the fourth and fifth examples, we focused on dynamic (Fig. 5a) and static (Fig. 5d) gesture recognition. Some gestures, such as those shown in Fig. 5d (static gestures 5, 13, 25, 26, 31 and 32), show high similarity, making it very challenging for current CV algorithms to distinguish. We employed the GlovePoseML model for each hand and added two FC layers to map finger-joint angles to the list of 50

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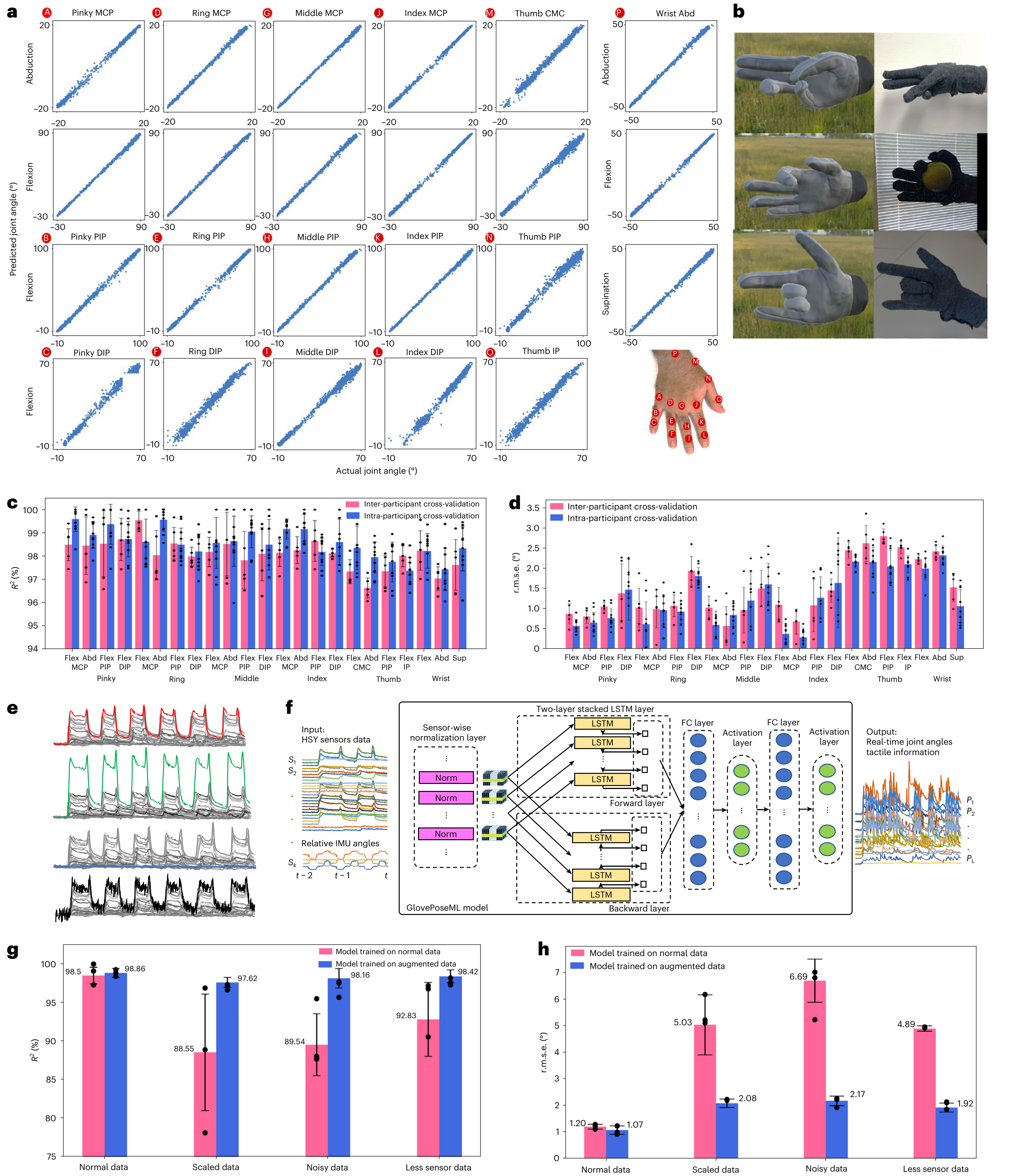

**Fig. 3 | ML and dynamic hand tracking.** **a**, Linear plot comparison of joint angles (A to P, shown in subset) estimated using GlovePoseML versus those measured by motion-capture camera and marker system. MCP, metacarpophalangeal; CMC, carpometacarpal; PIP, proximal interphalangeal; DIP, distal interphalangeal; IP, interphalangeal; Abd, abduction; Flex, flexion; Sup, supination. **b**, Comparison of visualized estimated hand pose (left) and photographs (right) of complex movements in normal conditions (top), when there is occlusion during grasping of a ball (middle) and low-light environment (bottom). **c**,**d**, Average accuracy results for different joints in terms of the goodness of fit $R^2$ (**c**) and r.m.s.e. (**d**). **e**, Scenarios for data augmentation. From top to bottom: original data, scaled data, data with fewer active sensors and noisy data. **f**, Overall architecture of tracking GlovePoseML model, showing the sensor $k$ response $S_k$, the normalization layer, two-layer stacked Bi-LSTM model, two FC layers, activation layers, time window $t$, and joint $L$ angle $P_L$. **g**,**h**, Comparison of average accuracy results of model trained using normal and augmented datasets in terms of goodness of fit $R^2$ (**g**) and r.m.s.e. (**h**). Bar charts in **c**, **d**, **g** and **h** show cross-validation averages with individual values represented by dots (tenfold for intra-participant and fivefold for inter-participant CV, respectively). Error bars indicate standard deviation.

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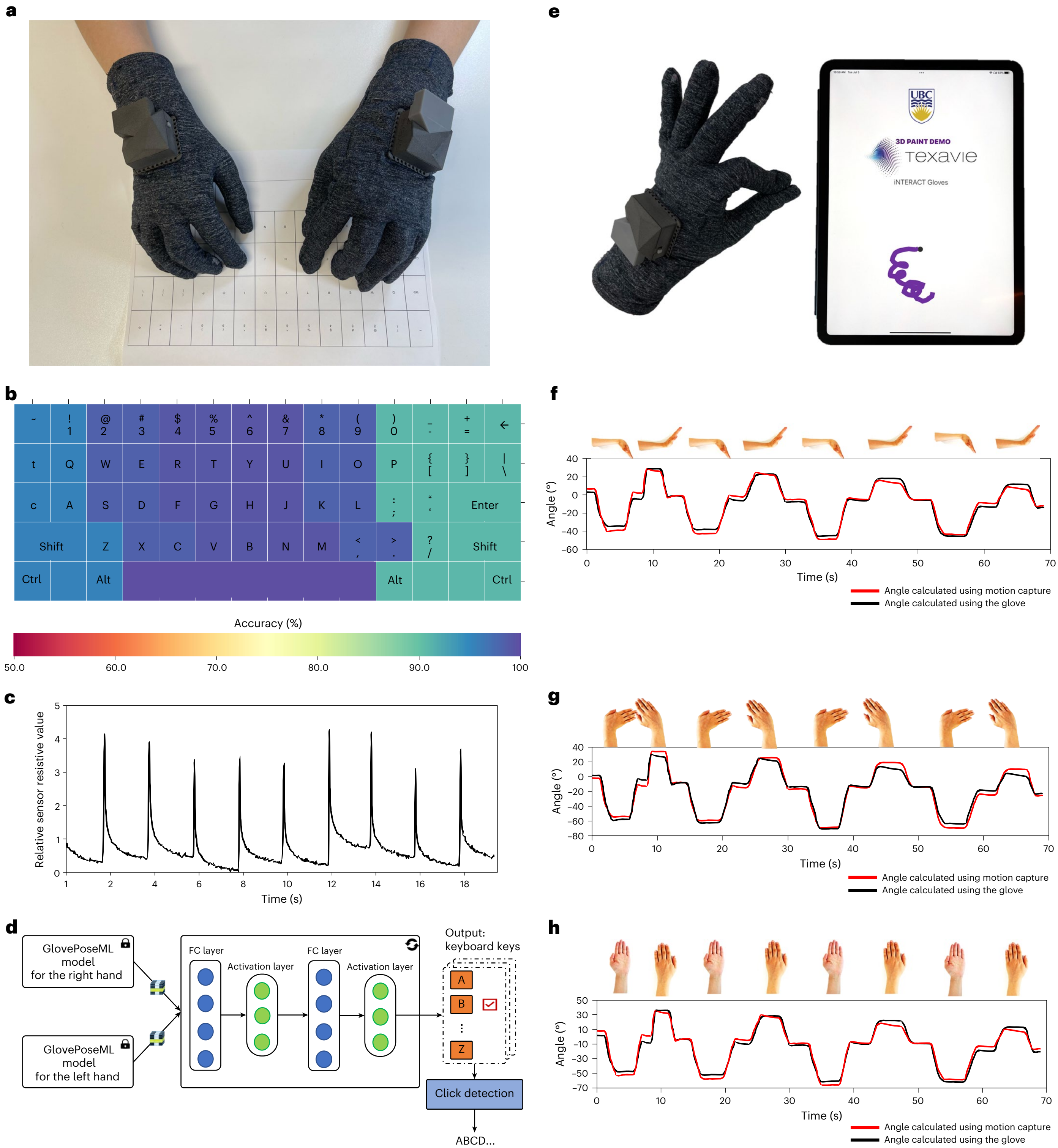


**Fig. 4 | Typing and drawing demos.** **a**, Photograph of user using a pair of wireless smart gloves for typing on a mock paper keyboard used for user's visual feedback. **b**, Colour-coded comparison of accuracy of detection of each key on the keyboard in ten-finger typing on the mock paper keyboard. **c**, Typical response of HSY sensor at the tip of a finger as it repeatedly taps and retracts the surface of the mock paper keyboard. **d**, Schematic of the two FC and activation layers that work with the GlovePoseML model for detection of typing. **e**, Illustration of a 3D in-air drawing application on an iPad based on two-finger pinch and wrist movement. **f**–**h**, Comparison between the estimated angle of the wrist using our smart glove and ML system and the gold-standard motion-capture system for tracking wrist flexion and extension (**f**), wrist abduction and adduction (**g**), and wrist supination and pronation (**h**).

dynamic gestures or 48 static gestures featuring complex finger and wrist poses, adapted from American Sign Language. We trained the added two layers while keeping the GlovePoseML models as is (Fig. 5j). We report inter- and intra-participant cross-validation results of 94.05% and 97.31% accuracy for detecting dynamic gestures (more details in Extended Data Fig. 4), and inter- and intra-participant cross-validation accuracy of 94.60% and 97.81%, respectively, for 48 static gestures (more details in Extended Data Fig. 5). To visualize the effectiveness

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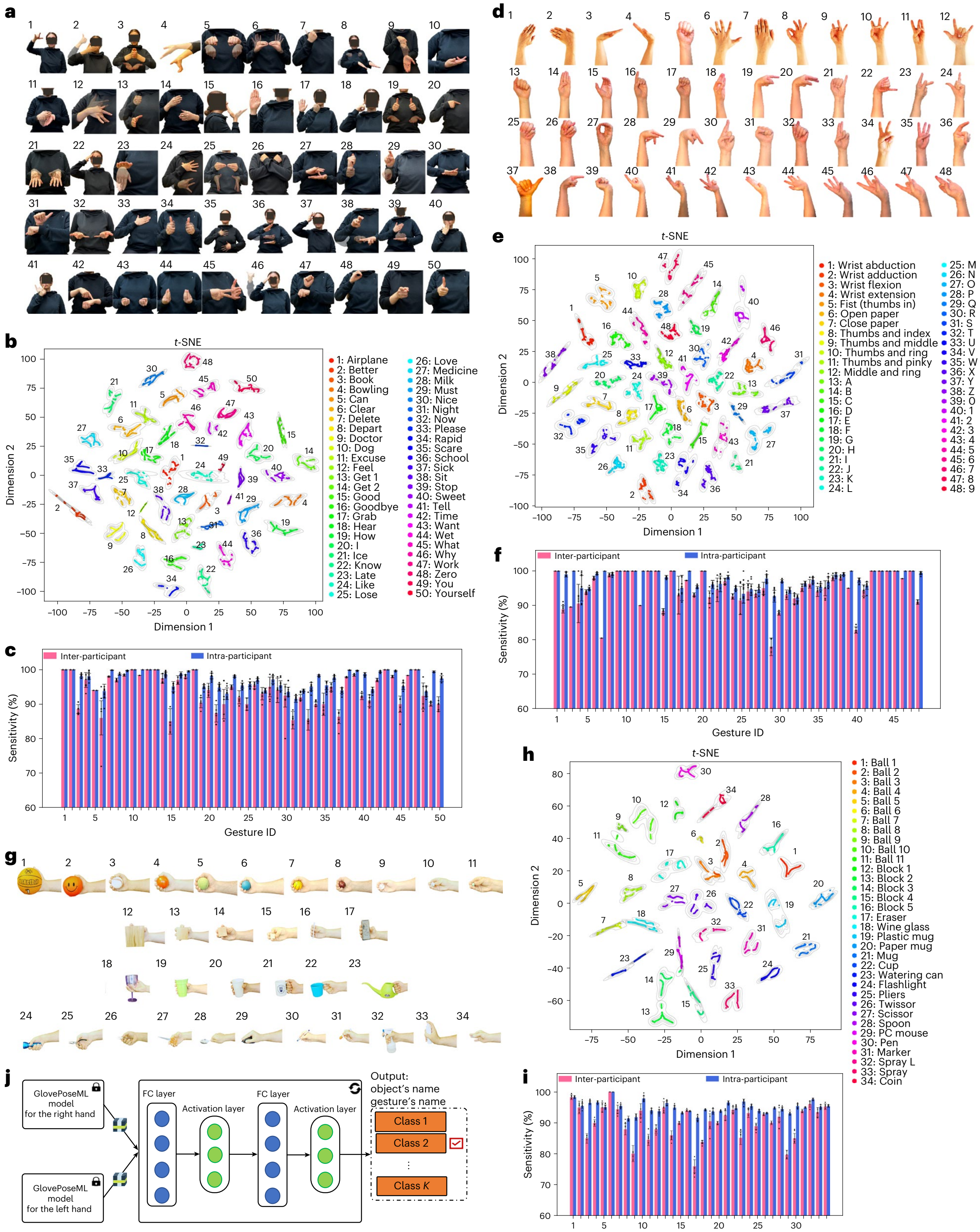


**Fig. 5 | Real-time dynamic and static gesture and object recognition.** **a**, Pictures of dynamic gestures used for hand-gesture recognition. **b**, Cluster distribution of dynamic gestures from GlovePoseML output layer between users. **c**, Sensitivity (%) results for dynamic gestures in inter- and intra-participant cross-validation. **d**, Pictures of static gestures used for hand-gesture recognition. **e**, Cluster distribution of static gestures from GlovePoseML output layer between users. **f**, Sensitivity (%) results for static gestures in inter- and intra-participant cross-validation. **g**, Pictures of objects used for object recognition from grasp form study. **h**, Cluster distribution of objects from GlovePoseML output layer between users. **i**, Sensitivity (%) results for different objects in inter- and intra-participant cross-validation. **j**, Schematic of the model used for gesture and object classification, showing FC and activation layers. Bar charts in **c**, **f** and **i** show cross-validation averages with individual values represented by dots (tenfold for intra-participant and fivefold for inter-participant CV, respectively). Error bars indicate standard deviation.

of our classification algorithm, we concatenated the output from the GlovePoseML models for both hands. We employed the *t*-distributed stochastic neighbour embedding (*t*-SNE) method to visualize the data derived from these concatenated outputs (Fig. 5b for dynamic and Fig. 5e for static gestures). Figure 5c,f shows the high sensitivity of our method for each gesture.

For the sixth application, we focused on detecting objects based on the participants' grasp patterns. We employed the GlovePoseML model and added two layers of FC neural network, mapping finger-joint angles to a list of 34 objects (Fig. 5g) with slight difference in shapes, weights, stiffness and grasp patterns. Here, the users are instructed to mimic a grasp pattern for each object using an instruction video. We trained these added two layers while keeping the core models as is (Fig. 5j). We report inter- and intra-participant cross-validation results of 90.20% and 95.02% accuracy in detecting the objects from an individual's grasp, respectively (Fig. 5e; more details in Extended Data Fig. 6). The algorithm makes use of different hand poses used for objects (for example, objects 11 and 34) as well as slight differences in pose and grasp pressure patterns (for example, objects 9, 10 and 11) for objects with very similar hand pose for accurate recognition. The distribution of clusters representing different objects realized by *t*-SNE for different participants in Fig. 5h, as well as object-wise sensitivity of our system reported in Fig. 5i demonstrates the effectiveness of our classification method.

## Conclusion

We report accurate, stretchable, washable and multi-modal smart textile gloves, with embedded stretchable HSYs, IMUs and interconnects, which can dynamically track movements of finger and hand joints and grasp forces during object interactions. The lightweight and insulated sensor yarns show high dynamic range in response to strains as low as 0.005% and as high as 155%, and low hysteresis and high stability during extensive use and washing cycles. Using ML algorithms, we demonstrate high accuracy for the smart gloves in estimating hand-joint angles with less than 1.21° and 1.45° average r.m.s.e. compared with the gold-standard motion-capture system for intra- and inter-participant cross-validation studies, respectively. A data augmentation technique is developed that enhances the robustness of the system to variations in the performance of the sensors, noise and fit to the participants. The reported smart gloves and ML system highlight the potential for high-accuracy tracking of dexterous hand movements and object interactions without limitations of field of view, occlusion and multi-user challenges associated with camera-based systems. We highlight the performance of our system by demonstrating real-time tracking of complex hand movements with minimal delay, as well as accurate detection of 48 static and 50 dynamic gestures adapted from American Sign Language, typing on a random surface such as a mock keyboard, and recognition of objects from grasp pose and forces. These results lay the foundation for learning realistic, dexterous human-hand movements and object interactions for creating new experiences in the metaverse, robotics control, tele-surgery, tele-rehabilitation and gaming applications, as well as potential for analysis of muscle function and grasp forces for assessment of patients with hand impairments. By extending the volume of data from different movements, interactions and gestures in various scenarios, increased depth and knowledge of hand dexterous functions will empower future realistic digital experiences and human–robot interactions.

## Methods

### Fabrication

The HSYs were fabricated using needleless electrospinning (NS LAB Nanospider, Elmarco) for deposition of NFs directly onto a core spandex polyurethane yarn (3,600 denier; weight in grams of 9,000 m of length) with a diameter of 0.5 mm (Crystal Tec) in a roll-to-roll fashion. Before electrospinning, the core yarn was cleaned using oxygen plasma (Tergio-plus) as well as acetone, isopropanol and deionized water. Polyacrylonitrile (Scientific Polymer Products) was dissolved in dimethylformamide (10 wt%, Fisher) and stirred at 60 °C for 24 h to form a homogeneous solution for electrospinning under a d.c. bias of 1.5 kV $cm^{-1}$. The core yarn is held between the bias wires of the electrospinning system and rotated around itself (60 rpm) using an electric motor to form helical coating of the NFs with an average diameter of 300 nm (standard deviation of 25 nm). After NF deposition, the yarns are coated with a 40–50 nm conformal layer of gold by plasma sputtering (Edward) to form a conductive HSY. Ag-coated nylon threads (Noble Biomaterials) were used as contact electrodes and knotted to the sensor yarn at desired locations over the length of the yarn and bound with silver paste (Pelco). The Ag-coated nylon thread (denier 100, linear resistance 0.8 Ω $cm^{-1}$) is a 3-ply yarn, the single plies are twisted at 16 twists per inch in the *S* direction and the 3 plies are twisted together at 14 twists per inch in the *Z* direction. The yarns were cured at 70 °C for 30 min to reach the optimal mechanical and electrical robustness. The sensor yarns are then encapsulated with PDMS (Dow Corning) by pouring and completely curing at 40 °C for 24 h using a tube-shaped mould to form a tubular all-around insulation.

The smart glove was prepared using a stretchable single jersey plated weft-knitted fabric (45% nylon, 45% polyester and 10% spandex), which was patterned and cut for desired glove patterns. The fabric has the following geometrical and mechanical properties (ASTM-D5035 Standard): stitch density, 2,142 loops per square inch; thickness, 0.61 mm; weight, 280 gm $m^{-2}$ GSM; breaking force, 320 N; breaking elongation, 110%; and elastic recovery, 94.8%. The sensor locations and their corresponding wiring were designed for the given size of the glove. A wiring bundle was made for each branch of the sensors routed for each finger (3–5 sensors in each branch) by twisting thin (34–38 AWG) insulated copper wires and mounting it on the fabric in a wavy form using a sewing machine to achieve the desired stretchability for the interconnects. The yarn sensors were placed and attached to the designated locations on the fabric corresponding to joint sensors and tips of the finger as shown in Fig. 1a using elastomer (Dragon Skin, Smooth-On). The fabric pieces with sensors are cured at room temperature for 30 min, and then the inside and outside fabrics are sewn together to form the final smart gloves. The interconnects are connected to a flexible printed circuit that is plugged into the main board and box. The fabrication process is shown in Extended Data Fig. 1a.

### Characterization of the HSYs

Tensile testing (INSTRON 5969) was performed to investigate the mechanical performance of the HSYs during stretching at strain >2% and pressing. The top end of the HSY sample was fixed on the system dynamometer to measure and control the force applied to the sensor, and the other end was anchored onto a steady holder. By adjusting the frequency and amplitude, the output under different strains was recorded. For compression studies, a 5 × 5 mm metal indenter was used to tap on the centre of sensors. Low-strain tensile testing (<2%) was conducted using a micro actuator (Zaber Technologies). The current and voltage measurements were acquired simultaneously by a Keysight B1500A semiconductor analyser.

To test the durability of the HSYs against daily laundry, the gloves were washed and dried using a typical laundry machine and detergents (Samsung front load, Samsung electric compact dryer, Purex liquid laundry detergent, Downy ultra fabric softener liquid). The electrical and mechanical properties of the gloves were monitored after each test.

### Electronic hardware and software

The movement data were acquired at a sampling rate of 20 Hz using a custom PCB (Texavie) containing multiplexers, pre-amplifiers and ADCs to measure 25 HSYs with a 12-bit resolution and two IMUs (Bosch BNO055) controlled by a centralized microprocessor (Nordic Semiconductor) with BLE connectivity. The power consumption details are included in Extended Data Fig. 2.

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The data collection gateway was implemented in a mobile application written in Swift 5 (Texavie), which collects data transmitted from each pair of gloves and stores the data in a database. It also sends a start–stop command through a WebSocket to a Python script running on a personal computer for synchronization with gold-standard motion-capture cameras (Optitrack). To compensate the shifting of the baseline resistance, we developed a dynamic baseline correction module in the data acquisition software to subtract the sensors' resistance values from their average value over a window of size $n = 400$ (20 s).

### Data collection and training

We collected data from five healthy right-handed participants (two female and three male participants) between the ages of 15 and 35 on both hands. In this study, we used one stretchable glove pair for all the participants, which provided a conformal fit for the participant during the entire data collection session. Informed consent was obtained from all participants (ethics obtained from UBC Clinical Research Ethics Board–H21-03021 entitled 'iGRASP:Phase 2 (Version1.0)'). Our data collection consisted of three main sections collected both from the gold-standard motion-capture system as well as the pair of proposed gloves system: (1) tracking hand- and wrist-joint angles in random movements or keyboard typing, (2) dynamic and static hand-gesture recognition, and (3) grasping different objects. Each section contains a list of movements that the participants were asked to follow using a prerecorded video supervising them to do a set of gestures.

**Articulated dynamic hand tracking.** We asked participants to follow a set of random finger and wrist movements. To mitigate task-related bias, participants were asked to follow a set of prerecorded videos. We recorded over 3,000,000 frames of data, including from the HSYs and IMUs for actual finger- and wrist-joint flexion and supination angles and from the motion-capture system. The wrist quaternion angle data were derived from the two IMUs and downsampled to have similar data rate to that of HSY data and then fed as input to GlovePoseML. The decision to fuse both modalities is due to their high correlation with joint angles and similar characteristics and computational efficiency, as treating the data from two sensors in different network branches did not show significant performance improvements.

We developed a recurrent neural network-based regression Bi-LSTM neural network architecture, which maps a 2 s window of sensor values to one set of hand-joint angles. We used the Adam optimizer with a learning rate of 0.0001. We trained our model for 100 epochs. We used a smooth L1 loss function with $\beta = 0.5$. More comprehensive results on the effect of customizing network architecture on joint-wise r.m.s.e. and $R^2$ can be found in Supplementary Figs. 10 and 11.

To enhance robustness, we augmented collected gloves data using three data transformations: (1) randomly masking one to three channels, (2) randomly adding Gaussian noise (with the mean of $\mu = 0$ and standard deviation of $\sigma = 0.06$) to one to three channels, and (3) scaling one to three channels by a random scalar between 0.5 to 1.5. Later, we trained our ML model in the multi-task learning setting. More details can be found in Supplementary Algorithm 1. More comprehensive results on the effect of noise level, and the number of masked or scaled sensors on GlovePoseML trained on normal data and augmented data, in terms of average r.m.s.e. and $R^2$ can be found in Supplementary Fig. 12. A demonstration video showcasing dynamic articulated tracking of finger movements can be found in Supplementary Video 1.

**Keyboard typing detection.** We asked the users to click on a single button of the keyboard. We developed a click-detection algorithm based on the HSY sensors at the tip of the fingers (more details in Supplementary Algorithm 2). We use two FC layers to map the output of the GlovePoseML model to ten keys that the fingers manoeuvre over using the ten-finger typing method. The keyboard was marked using four motion-capture markers placed on the corners of the printed keyboard. This was used to calibrate key locations as our ground truth. At the start of data collection, we asked the participants to pose their hands over the keyboard and stay in this position for 10 s as the resting position in the ten-finger typing method. Then, we asked them to type a paragraph containing 100 characters on a printed keyboard for training of the two output layers. We collected five different trials and performed fivefold cross-validation. A demonstration video showcasing typing on a mock paper keyboard can be found in Supplementary Video 2.

**3D drawing.** We developed a Mobile interface app using Swift 5. We developed the cursor's location on the screen based on the relative angle of the two IMUs' quaternion values. Users could choose different colours by pinching the thumb using different fingers detected by using a touch detection algorithm similar to click detection from the HSYs at the tip of the fingers. A demonstration video showcasing 3D drawing in air can be found in Supplementary Video 3.

**Dynamic and static hand-gesture recognition.** We recorded a set of videos each containing one gesture and rest position. The gesture was shown for 5 s and users had 5 s to rest between each trial. We collected 20 sets of data for each gesture in total. We trained the last two output layers while freezing the GlovePoseML models. Inter-session cross-validation results can be found in Supplementary Figs. 1–3 for dynamic gesture recognition, and Supplementary Figs. 4–6 for static gesture recognition. Also, demonstration videos showcasing static and dynamic hand-gesture recognition can be found in Supplementary Videos 4 and 5.

**Object detection from the grasp.** We recorded a set of videos containing grasping one object and rest position. Grasps were shown for 5 s and users had 5 s to rest between each trial. We collected 20 sets for each gesture in total. We created a tensor mapping window for each dataset over sensor data to the list of objects the user grasped. We combined all tensors and shuffled data for each participant to create a training dataset. We employed the hand-joint-angle estimation architecture we trained on earlier. We added two FC neural networks mapping hand-joint angles to a list of objects. During the training phase, we updated the last two layers while freezing the rest of the network. More results can be found in Supplementary Figs. 7–9. More details about the properties of objects used in this study can be found in Supplementary Table 2. A demonstration video showcasing the object-detection algorithm can be found in Supplementary Video 6.

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

## Data availability

The data supporting this study's findings are available from project page, containing detailed explanations of all the datasets (https://feel.ece.ubc.ca/SmartTextileGlove/) as well as a direct link to a Google Drive Repository (https://drive.google.com/drive/folders/1HWjG_6Y2G7XNEeI19Aids0g-dcufncGJ?usp=share_link) where the datasets can be downloaded. Source data are provided with this paper.

## Code availability

The codes supporting this study's findings are available from https://github.com/arvintashakori/SmartTextileGlove ref. 53.

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## Author contributions

A.T. and P.S. developed the system model and implemented the learning algorithm, iOS Mobile application, data pipeline, PC-based data acquisition software, Unity application and firmware parts. P.S., Z.J. and S.S. designed the yarn-based strain sensors. Z.J., A.S., S.S., H.N., K.L. and P.S. developed the hardware and fabricated gloves and sensors. A.T. performed the experiments and analysis with help and input from others. C.N. helped with the PCB box design and fabrication. P.S., A.S., J.J.E., C.-l.Y. and Z.J.W. oversaw the project. All authors contributed to writing of the paper and analysis of results.

## Competing interests

P.S., A.T., Z.J., C.N., A.S., S.S. and H.N. have filed a patent based on this work under the US provisional patent application no. 63/422,867. The other authors declare no competing interests.

## Additional information

**Extended data** is available for this paper at https://doi.org/10.1038/s42256-023-00780-9.

**Supplementary information** The online version contains supplementary material available at https://doi.org/10.1038/s42256-023-00780-9.

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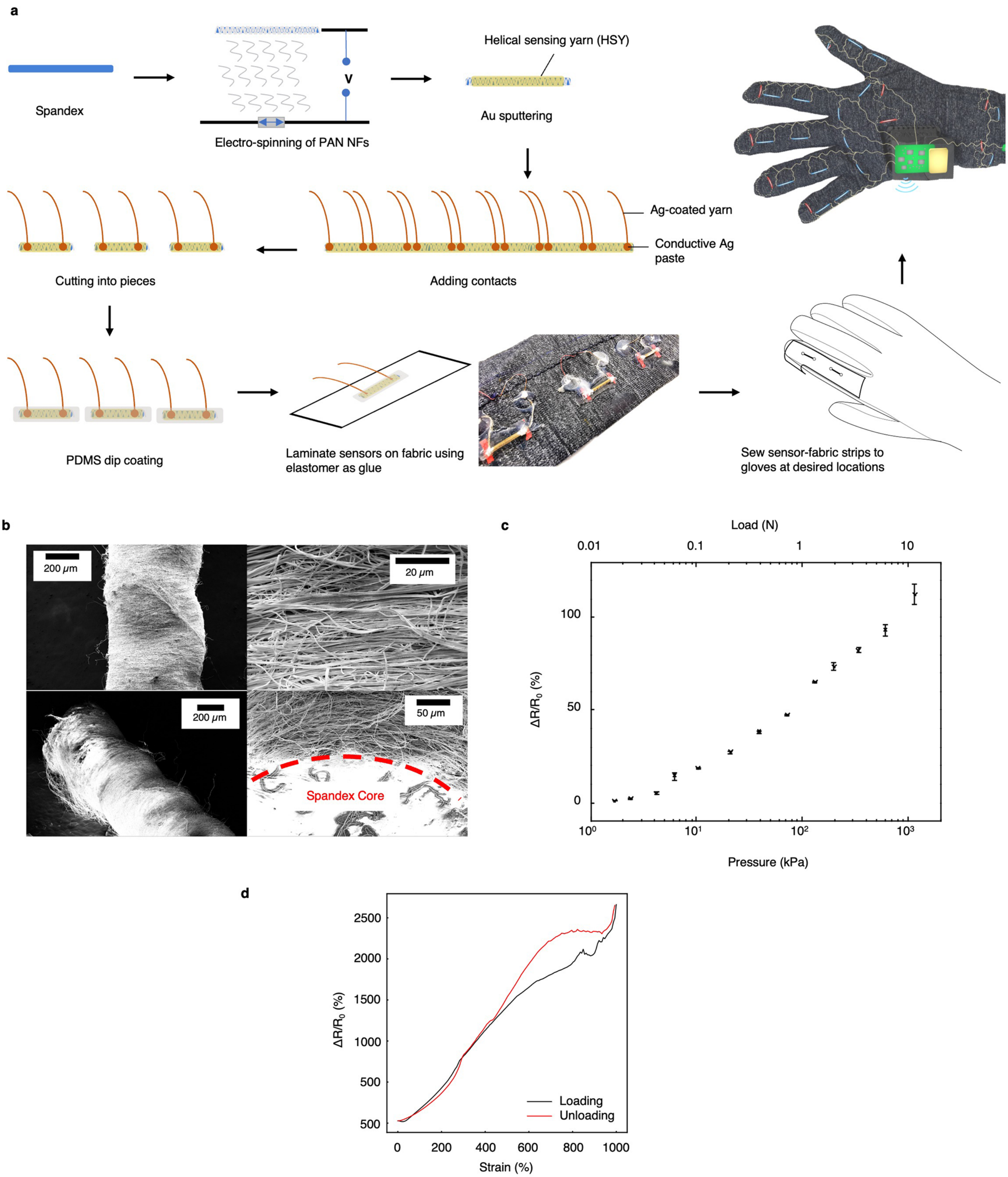


**Extended Data Fig. 1 | Fabrication and characteristics of HSYs. a**, Fabrication process of HSYs and gloves. **b**, SEM images of the yarn sensors before PDMS coating. **c**, Sensitivity of HSY resistance to various compressive pressure values applied at a frequency of 1 Hz. A metal indenter with the size of $5mm \times 5mm$ were used to apply pressure normal to the fabric. (The data points present the mean values ± standard deviation of 20 samples.) **d**, The strain sensitivity of our insulated yarn sensors made from optimized composites of carbon particles with highly stretchable elastomers, demonstrating high stretchability up to 1,000 %, but showing more hysteresis for > 500 % stretch and slower responsiveness due to the softer nature of these sensors in comparison to HSYs. This highlights the superior performance of HSYs for the proposed smart glove real-time applications with less than 120 % maximum stretch for the fabric.

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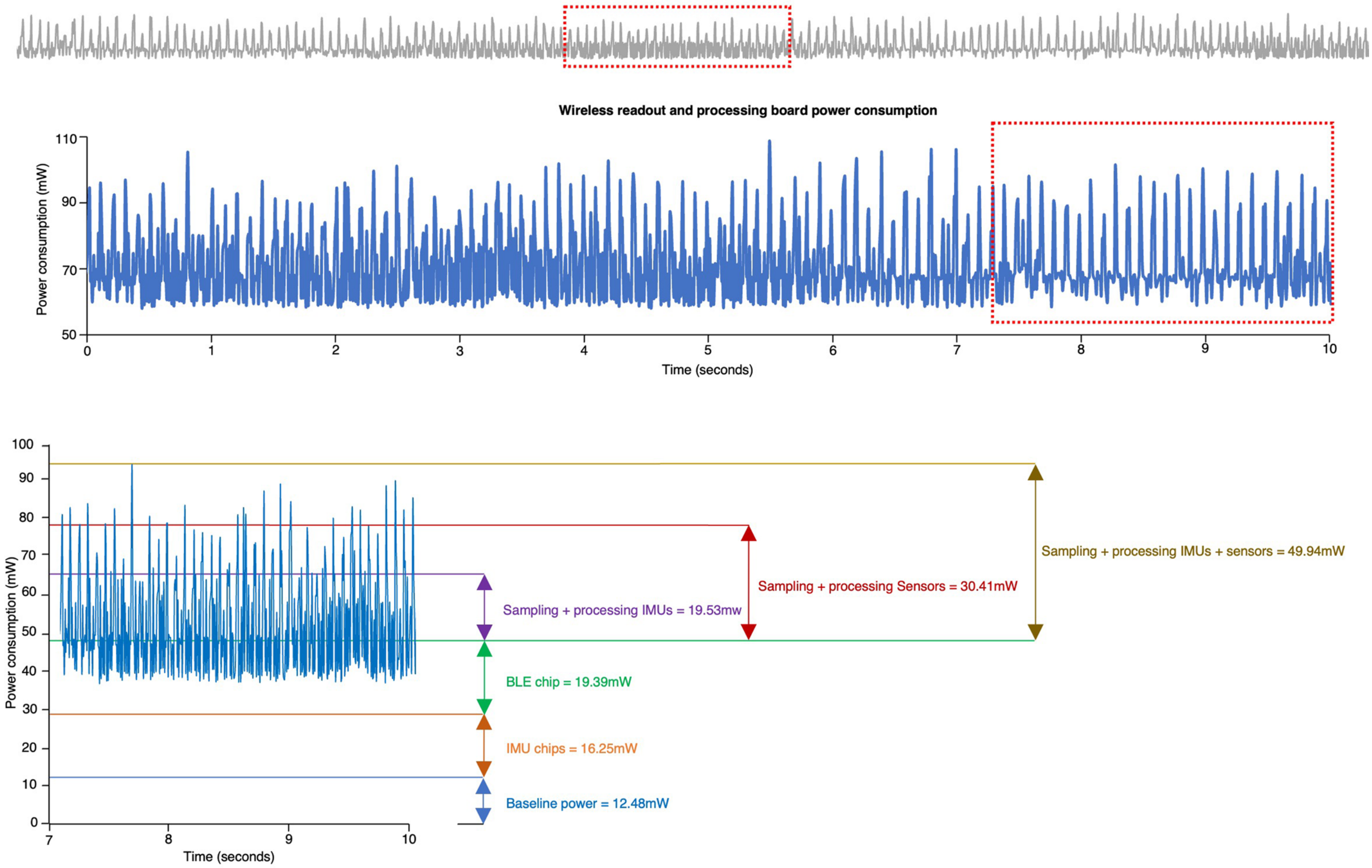
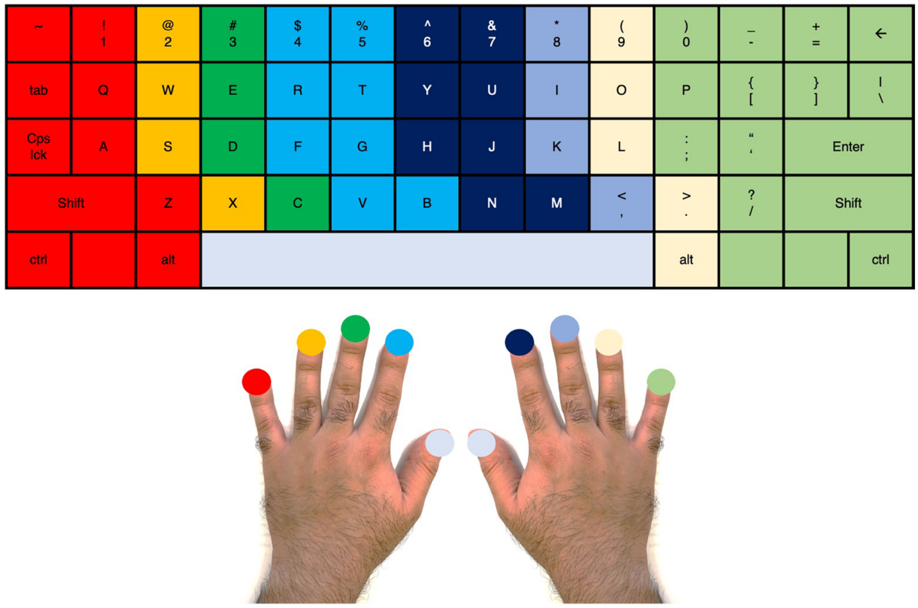



**Extended Data Fig. 2 | Power consumption.** Custom-made wireless board power consumption breakdown for different components including BLE chip, IMU chips, and all HSYs.




| Finger | Accuracy(%) |
|---|---|
| Right pinky | 91.01% |
| Right ring | 98.32% |
| Right middle | 98.14% |
| Right index | 99.37% |
| Right thumb | 100% |
| Left thumb | 100% |
| Left index | 99.14% |
| Left middle | 98.29% |
| Left ring | 98.38% |
| Left pinky | 95.34% |
| **Average** | **97.8%** |

**Extended Data Fig. 3 | Keyboard typing detection.** Inter-session cross-validation accuracy results for the keyboard typing detection algorithm.

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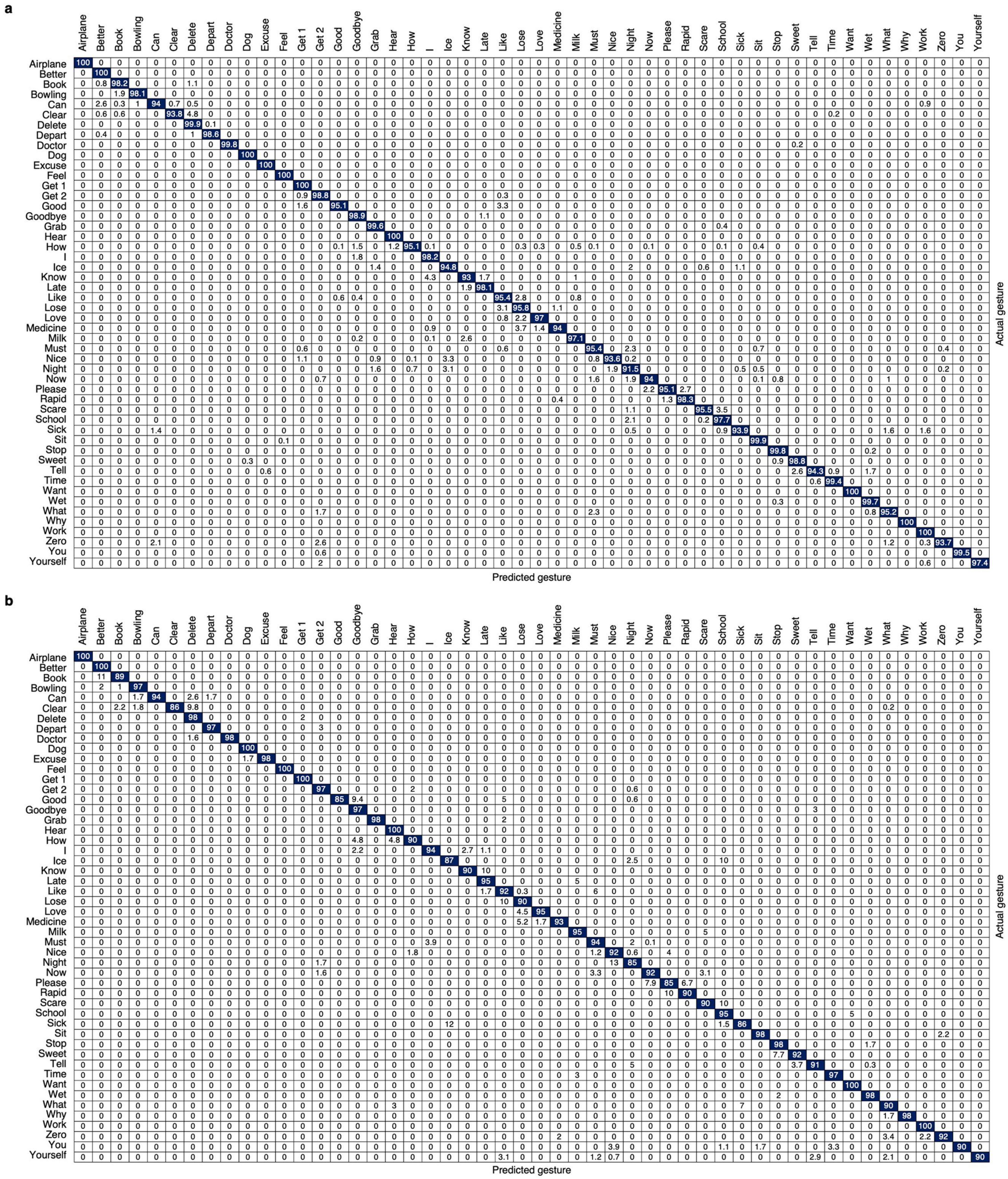
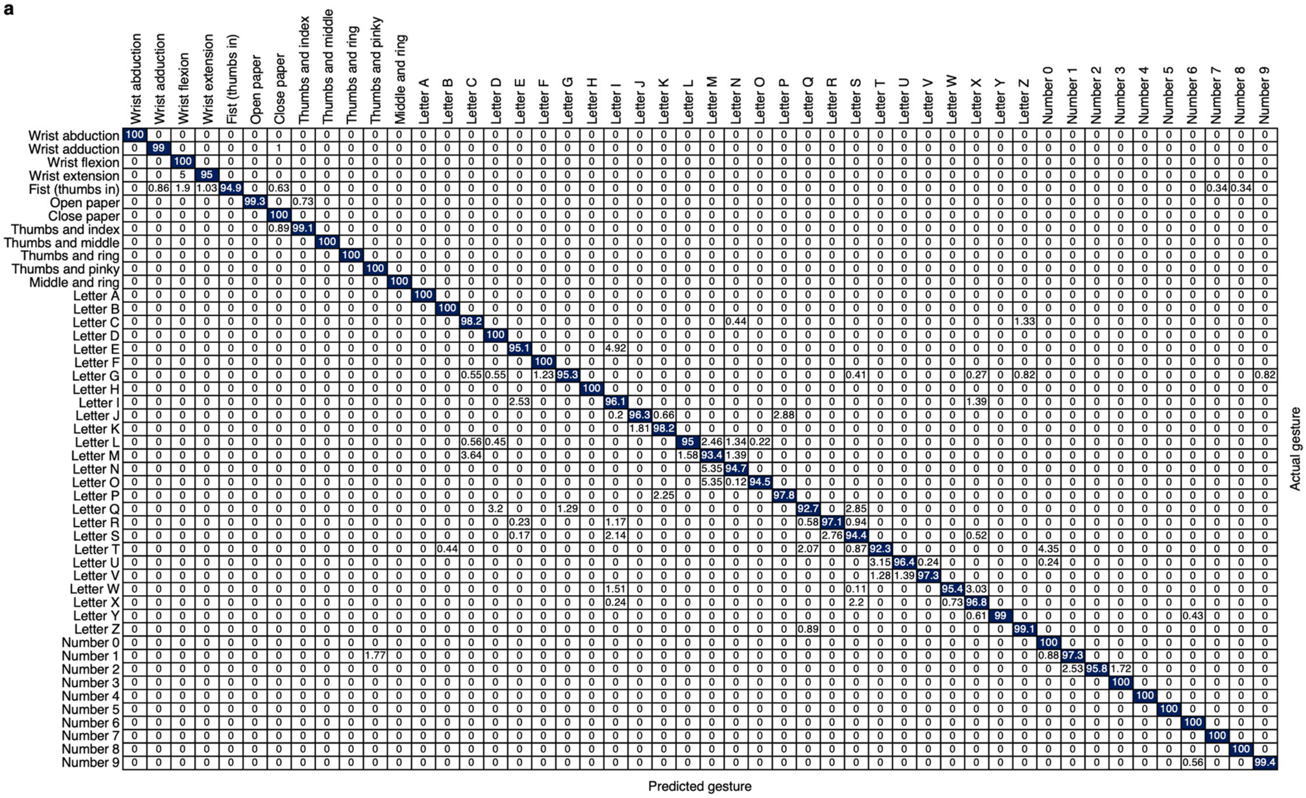
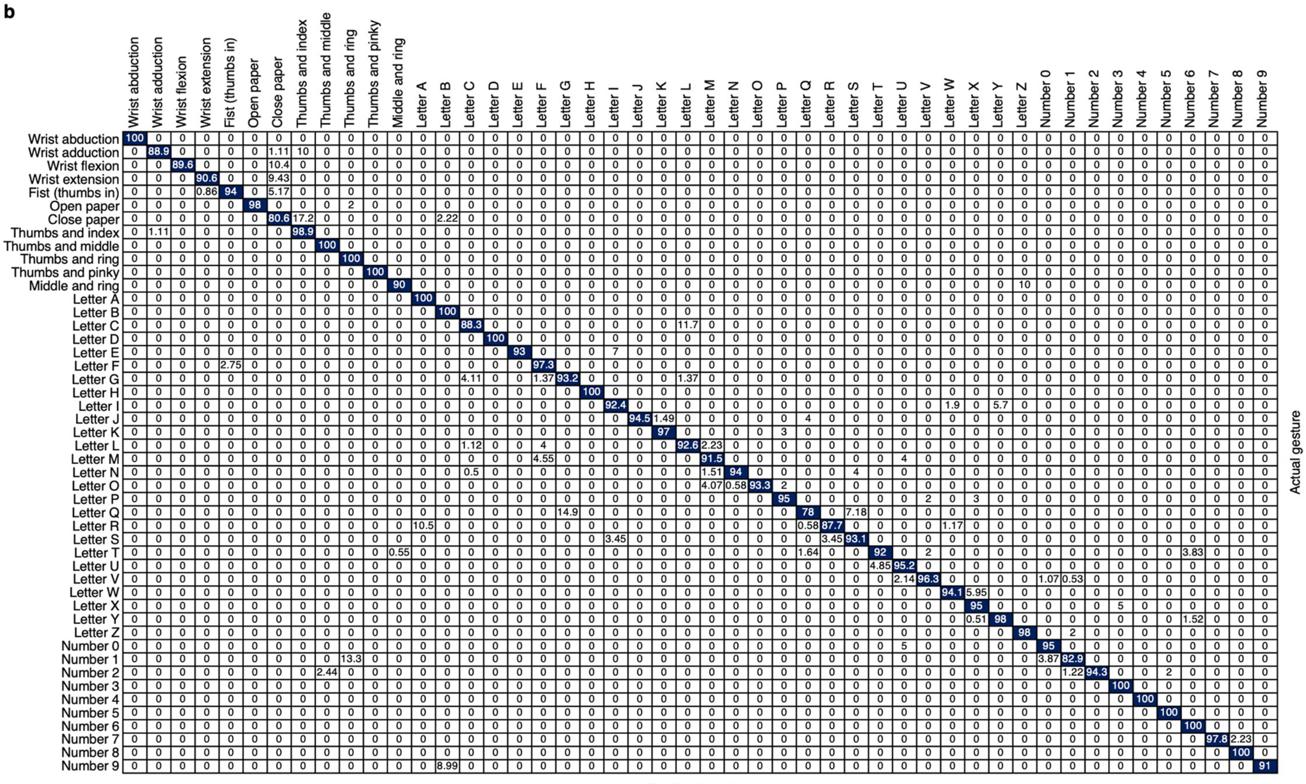



**Extended Data Fig. 4 | Dynamic gesture recognition.** Confusion matrix for **a**, intra-subject (accuracy: 97.31%), and **b**, inter-subject cross-validation (accuracy: 94.05%).

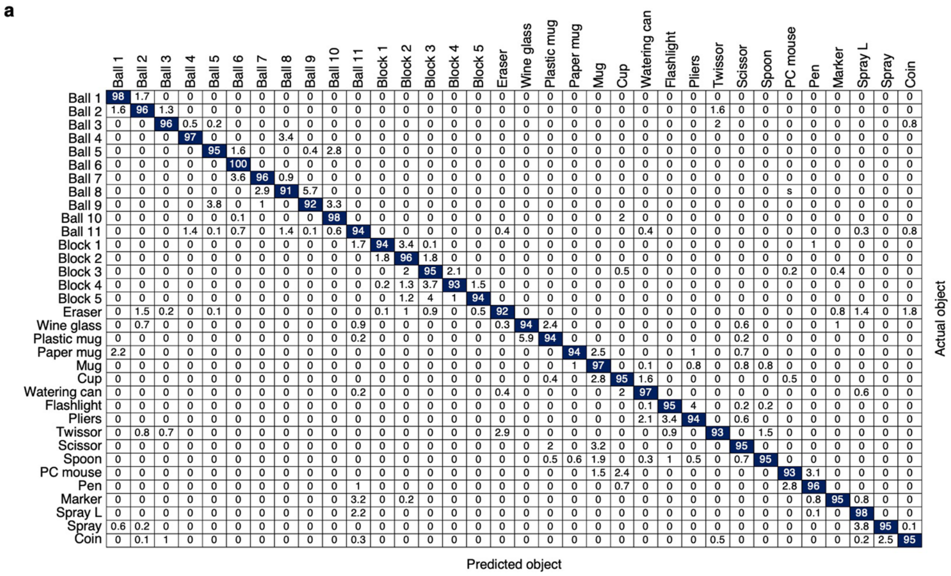
a

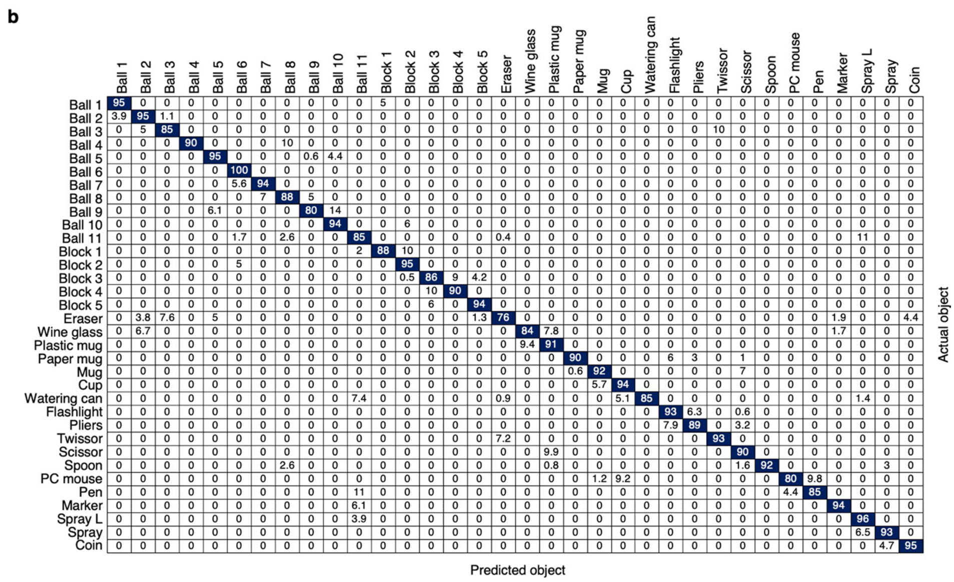



b




**Extended Data Fig. 5 | Static gesture recognition.** Confusion matrix for **a**, intra-subject (accuracy: 97.81 %), and **b**, inter-subject cross-validation (accuracy: 94.60 %).




**Extended Data Fig. 6 | Object detection.** Confusion matrix for **a**, intra-subject (accuracy: 95.02 %), and **b**, inter-subject cross-validation (accuracy: 90.20 %).

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# Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

## Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

| n/a | Confirmed | |
|---|---|---|
| ☐ | ☒ | The exact sample size ($n$) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ | A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☒ | ☐ | The statistical test(s) used AND whether they are one- or two-sided<br>*Only common tests should be described solely by name; describe more complex techniques in the Methods section.* |
| ☒ | ☐ | A description of all covariates tested |
| ☒ | ☐ | A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ | A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☒ | ☐ | For null hypothesis testing, the test statistic (e.g. $F$, $t$, $r$) with confidence intervals, effect sizes, degrees of freedom and $P$ value noted<br>*Give P values as exact values whenever suitable.* |
| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☒ | ☐ | Estimates of effect sizes (e.g. Cohen's $d$, Pearson's $r$), indicating how they were calculated |

*Our web collection on statistics for biologists contains articles on many of the points above.*

## Software and code

Policy information about availability of computer code

| | |
|---|---|
| Data collection | Our data collection consisted of three main sections collected both from the commercial gold standard motion capture system. Optitrack, as well as the pair of proposed gloves system: 1) tracking hand and wrist joint angles in random movements or keyboard typing, 2) dynamic and static hand gesture recognition, and 3) grasping different objects. Each section contains a list of movements that the subjects were asked to follow using a prerecorded video supervising them to do a set of gestures. Data from proposed glove collected using a custom-made wireless readout board transmitting data to a PC hosting a custom application developed in Python 3.8.1. The software simultaneously collect data from motion capture as well. |
| Data analysis | We performed all data visualization, signal processing methods, and analysis using customized codes developed based on publicly available Python packages such as Seaborn, Numpy, Pandas, and PyTorch. |

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

## Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

All data supporting findings in this study are available here: https://feel.ece.ubc.ca/smarttextileglove/

## Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

| | |
|---|---|
| Reporting on sex and gender | We collected data from two female and three male subjects |
| Population characteristics | We collected data from five healthy right-handed participants between the age of 15 to 35 on both hands. |
| Recruitment | We recruited subjects from our research group. |
| Ethics oversight | Ethics obtained from UBC Clinical Research Ethics Board - H21-03021 entitled "iGRASP:Phase 2 (Version1.0)" |

Note that full information on the approval of the study protocol must also be provided in the manuscript.

# Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

# Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

| | |
|---|---|
| Sample size | We collected data from two female and three male subjects between the age of 15 to 35 on both hands. |
| Data exclusions | No data excluded from this study. |
| Replication | Experiments were performed using different devices made with the same materials. Each experiment was performed five times for each subjects. all attempts at replication were successful. |
| Randomization | In this study as the experiments was specific to each participants, randomizations was not relevant. |
| Blinding | Since there was no subjective parameters blinding was not relevant to this study. |

# Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

## Materials & experimental systems

| n/a | Involved in the study |
|---|---|
| ☒ | ☐ Antibodies |
| ☒ | ☐ Eukaryotic cell lines |
| ☒ | ☐ Palaeontology and archaeology |
| ☒ | ☐ Animals and other organisms |
| ☒ | ☐ Clinical data |
| ☒ | ☐ Dual use research of concern |

## Methods

| n/a | Involved in the study |
|---|---|
| ☒ | ☐ ChIP-seq |
| ☒ | ☐ Flow cytometry |
| ☒ | ☐ MRI-based neuroimaging |